

QUANTUM KNOWLEDGE GRAPH

An edge may appear both **valid** and **invalid** until evaluated under a specific context.



Drug A

$$|\psi\rangle = \alpha |\text{valid}\rangle + \beta |\text{invalid}\rangle$$

Superposition
(context-dependent)



Disease B

Evaluated under
a specific context

Patient 1

Context

- ✓ eGFR ≥ 60
- ✓ No renal failure
- ✓ No allergy
- ...



— Observed state —

Valid
(EXIST)

Patient 2

Context

- ✗ eGFR < 30
- ✗ Renal failure
- ✗ Allergy to A
- ...



— Observed state —

Invalid
(NOT EXIST)



Relations are functions whose validity is defined only under context.